

◆ 16.4 Graph Structure and Network Coupling

16.4.1 Graph Definition

$$L_k = D_k - A_k$$

$$(L_k \Psi_k)_i = \sum_j a_{ij}^k (\Psi_i - \Psi_j)$$

where:

- $a_{ij}^k \geq 0$ are edge weights
- $a_{ij}^k = a_{ji}^k$ (undirected graph)

Thus:

$$L_k \succeq 0$$

16.4.2 Spectral Properties

$$0 < \lambda_2(L_k) \leq \lambda_{\max}(L_k) \leq \Lambda$$

This ensures:

- graph connectivity ($\lambda_2 > 0$)
 - bounded spectral norm
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16.4.3 Adaptive Coupling Law

$$\begin{aligned} w_{ij}^{k+1} &= \\ &w_{ij}^k \\ &+ \\ &dt \left(\right. \\ &-\kappa \|\Psi_i - \Psi_j\|^2 \\ &+ \beta, PAC_i, PAC_j \\ &\left. \right) \end{aligned}$$

with constraints:

$$w_{ij} \in [w_{\min}, w_{\max}]$$

16.4.4 Weight Normalization

$$\begin{aligned} a_{ij} &= \\ &\exp\big(\mathrm{clip}(w_{ij}, w_{\min}, w_{\max}) \\ &\big) \end{aligned}$$

ensuring:

- bounded weights
 - numerical stability (no overflow)
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16.4.5 Network Injection into Dynamics

$$\dot{\mathbf{F}} \supset -\mathbf{L}_k \Psi_k$$

Thus:

the network structure directly influences system evolution.

16.4.6 Dissipativity of Coupling

$$\begin{aligned} & \langle \Psi, \mathbf{L} \Psi \rangle \\ &= \\ & \sum_{i,j} a_{ij} \|\Psi_i - \Psi_j\|^2 \geq 0 \end{aligned}$$

Thus:

the coupling is dissipative and promotes consensus.

16.4.7 Computational Complexity

$$\text{Time} = O(E), \quad \text{Memory} = O(N + E)$$

where E is the number of edges and N is the number of nodes.

Thus:

the framework is scalable to large-scale systems.

16.4.8 Time-Varying Graphs

$\lambda_2(L(t)) > 0, \quad \forall t$

Thus:

stability is preserved under time-varying network topology.

16.4.9 Integration with Previous Sections

- From Section 16.2:

$F \supseteq L \cap \Psi$ (network enters the dynamics)

- From Section 16.3:

L contributes to Lyapunov dissipation

Thus:

the network is an integral component of system stability.

Closure Statement (Part IV)

\boxed{
\text{The system achieves stable, adaptive, and
coherent multi-agent coupling.}
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